

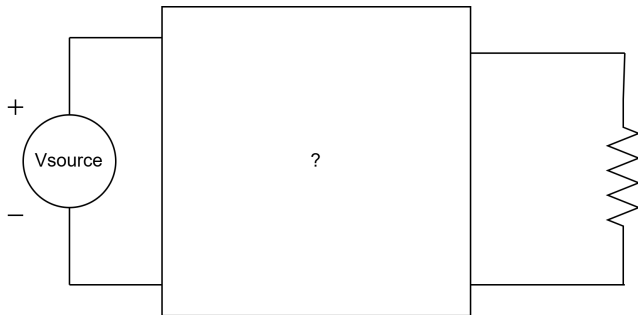
Power Electronics — Chapter 1

DC-DC Converters

POWER_E_466

Buck Converter: Intro

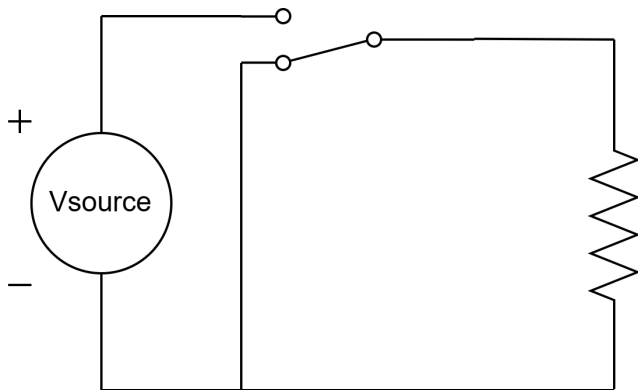
The buck converter is a DC-DC converter that steps down voltage ($V_{out} < V_{in}$).



Buck Converter: Switching

Start from a black-box model: Given $V_i = 50V$, our goal $\langle V_o \rangle = 20V$.

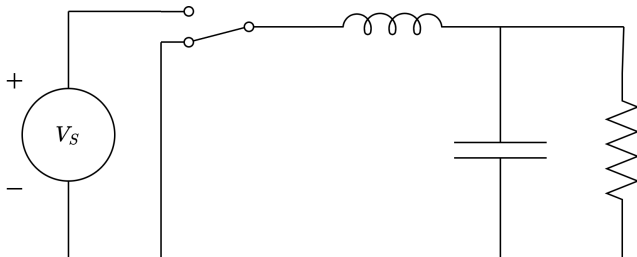
We start with just simple on-off switching.



We'll get PWM with average output voltage of 20V

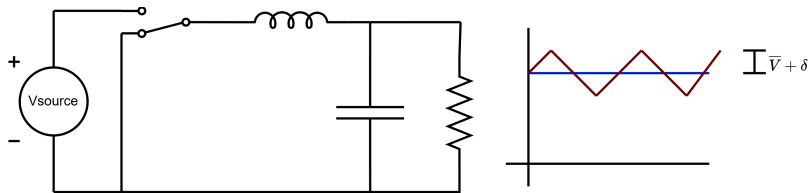
Buck Converter: Filter Added

From Fourier we know that DC offset have frequency at 0 hz.
($f = 0$ hz) To gain DC offset we used LC filter circuit which lossless in ideal.



Buck Converter: Small Ripple Assumption

For an easier circuits analysis we assume the filter provide small ripple voltage. ($\delta \approx 0$)



We can assume that V_o constant DC offset.

Buck Converter: Inductor and capacitor averaging identities

For an inductor,

$$v = L \frac{di}{dt} \iff i = \frac{1}{L} \int v dt,$$

so if i is periodic in steady state, the average inductor voltage must be zero:

$$\langle v_L \rangle = 0 \quad (\text{volt-second balance}).$$

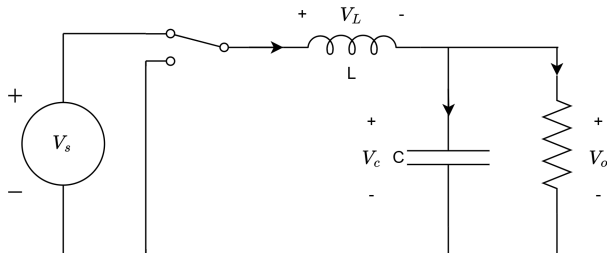
For a capacitor,

$$i = C \frac{dv}{dt} \iff v = \frac{1}{C} \int i dt,$$

so if v is periodic in steady state, the average capacitor current must be zero:

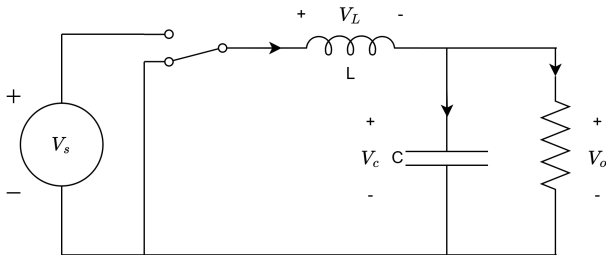
$$\langle i_C \rangle = 0 \quad (\text{current-second/charge balance}).$$

Buck Converter: Steady-State Switching On



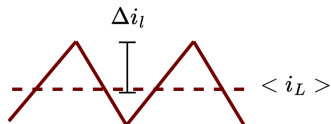
$$\frac{di_L}{dt} = \frac{V}{L} = \frac{V_s - V_o}{L}; \Delta t = DT; \Delta i_{L\text{-on}} = DT \cdot \frac{V_s - V_o}{L}$$

Buck Converter: Steady-State Switching Off



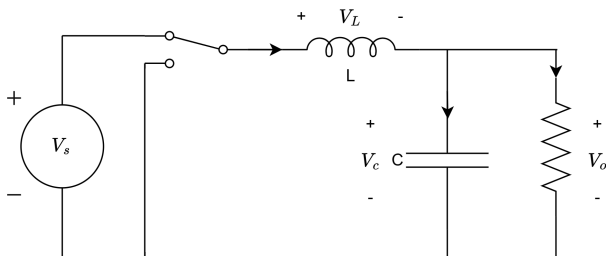
$$\frac{di_L}{dt} = \frac{V}{L} = \frac{V_o}{L}; \Delta t = (1 - D)T; \Delta i_{L\text{-off}} = (1 - D)T \cdot \frac{V_o}{L}$$

Buck Converter: Steady-State Inductor current



Buck Converter: Transient Switching On (1)

For easier analysis we use discrete domain to analyze. $V_s = 50V$
 $V_o = 0V$ $D = 0.5$ $L = 10mH$ $T = 1ms$

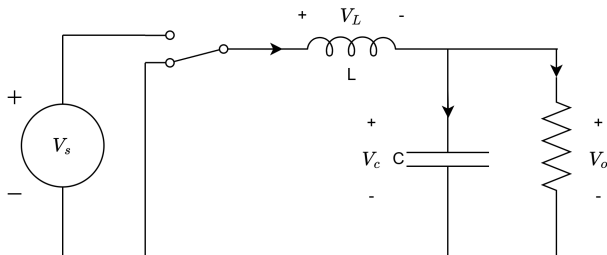


$$\Delta i_{L-on} = DT \cdot \frac{V_s - V_o}{L} = 5A; i_L = 5A$$

$$i_c = i_l - i_o = i_l - \frac{V_o}{R} = 5A$$

Buck Converter: Transient Switching Off (1)

$$V_s = 50V \quad V_o = 0V \quad D = 0.5 \quad L = 10\text{mH} \quad T = 1\text{ms}$$

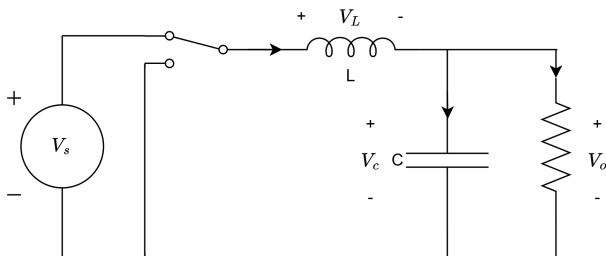


$$\Delta i_{L\text{-off}} = (1 - D)T \cdot \frac{V_o}{L} = 0A; i_L = 5A$$

$$i_c = i_l - i_o = i_l - \frac{V_o}{R} = 5A$$

Buck Converter: Transient Switching On (2)

For easier analysis we use discrete domain to analyze. $V_s = 50\text{V}$
 $V_o = 2.5\text{V}$ $D = 0.5$ $L = 10\text{mH}$ $T = 1\text{ms}$

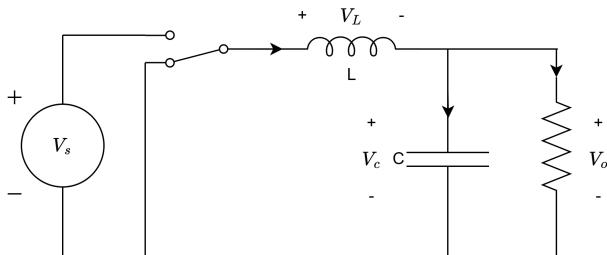


$$\Delta i_{L-\text{on}} = DT \cdot \frac{V_s - V_o}{L} = 4.75\text{A}; i_L = 9.75\text{A}$$

$$i_c = i_l - i_o = i_l - \frac{V_o}{R}$$

Buck Converter: Transient Switching Off (2)

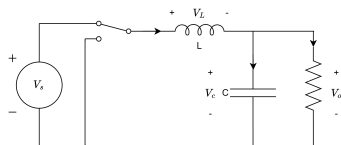
$$V_s = 50V \quad V_o = 2.5V \quad D = 0.5 \quad L = 10mH \quad T = 1ms$$



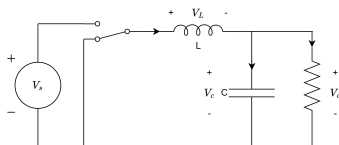
$$\Delta i_{L\text{-off}} = (1 - D)T \cdot \frac{V_o}{L} = 0A; i_L = 5A$$

$$i_c = i_l - i_o = i_l - \frac{V_o}{R}$$

Buck Converter: Summary



$$\Delta i_{L\text{-on}} = DT \cdot \frac{V_s - V_o}{L}$$



$$\Delta i_{L\text{-off}} = (1 - D)T \cdot \frac{V_o}{L}$$